

Data Preparation Session 7

Exercises

Exercise 7.1

- a) Read the `mexicanweather.csv` dataset and make it tidy.
- b) Separate the year, the month, and the day into new columns using `lubridate` functions.
- c) Look at the `station` column. What can you propose based on unique values?

Exercise 7.2

- a) Store your birth time in an object `mybirth`. Pay attention to the right time zone.
- b) Check if you were born in a leap year, or daylight-saving time.
- c) What day of the week?
- d) Calculate your age in duration and period. Explain the difference between them.
- e) Calculate when you will be 20,000 days old.
- f) What will be the time in Auckland then?

Exercise 7.3. (revision)

- a) Load the 'Survey Data For My Best Exam 2024.txt' file in R, using functions in the `tidyverse` library. Do NOT rename the file and do NOT edit it BEFORE loading.
- b) Detect and remove empty columns WITHOUT the `janitor` package.
- c) Rename 'Timestamp' to 'time' and transform it to the right data format. Check with a code if the transformation was successful and you don't have any NAs.
- d) What is the hour just now in El Salvador? Store it in a variable `EShour`.
- e) Look for outliers in the 'Annual Salary'. Remove the people with the 3 most extreme outliers.
Hint: to remove spaces from the annual salary, you can use the following code.

```
survey$`Annual salary` <- str_replace_all(survey$`Annual salary`, " ", "")
```
- f) Look for outliers in the 'Years of experience in field' and 'Overall years of professional experience' columns. Explain in a comment what is wrong with them.
- g) Clean the 'Years of experience in field' and 'Overall years of professional experience' columns and conclude on the relationship between the two variables (no statistical tools are needed).